

USP26 functions as a negative regulator of cellular reprogramming by stabilising PRC1 complex components

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SITUATIONAL CONTEXT HOOK

In mouse embryonic fibroblasts (MEFs) undergoing Oct4-Sox2-Klf4-cMyc (OSKM)-mediated cellular reprogramming and embryonic stem cells (ESCs) undergoing differentiation, the deubiquitinase USP26 acts as a potent negative regulator of pluripotency. Co-immunoprecipitation assays in HEK293T cells and differentiating ESCs demonstrate that USP26 physically interacts with canonical PRC1.2 complex components, specifically CBX4 and CBX6, in a RING1A-independent manner. USP26 enzymatic activity removes K48-linked polyubiquitin chains from CBX4 and CBX6, protecting them from 26S proteasomal degradation, whereas catalytic mutant USP26-C304S fails to deubiquitinate them. Accumulated CBX4 and CBX6 recruit PRC1 complexes to pluripotency gene promoters, promoting H2AK119 monoubiquitination and transcriptional repression of Sox2 and Nanog. Knockdown of Usp26 increases alkaline phosphatase-positive iPSC colony generation fourfold (~400 vs ~100 colonies) and upregulates Sox2 and Nanog expression.

KEY FINDINGS (5)

shRNA knockdown of Usp26 in MEFs enhances reprogramming efficiency fourfold, yielding approximately 400 AP+ iPSC colonies.

protein_pair USP26 ↔ NANOG
evidence ~400 AP+ colonies vs ~100 in control shRNA (quantitative)
context Dox-induced OSKM MEF reprogramming at day 12
confidence 0.95 · significant: True
source_span Page 3, Para 1

USP26 physically binds to PRC1.2 components CBX4 and CBX6 independently of the presence of RING1A.

protein_pair USP26 ↔ CBX4
evidence Co-IP in WT and RING1A KO 293T cells (qualitative)
context Anti-FLAG IP in transfected 293T cells
confidence 0.95 · significant: False
source_span Page 5, Para 1

USP26 deubiquitinase activity specifically removes K48-linked polyubiquitination from CBX4 and CBX6 via active-site Cys304.

protein_pair USP26 ↔ CBX6
evidence Loss of K48 deubiquitination in C304S and Δ295-312 mutants (qualitative)
context In vitro deubiquitination and IP in 293T cells
residues Cys304
confidence 0.95 · significant: False
source_span Page 5, Para 1

Accumulated CBX4 and CBX6 bind to Sox2, Nanog, and Cbx7 promoters to repress their transcription.

protein_pair CBX4 ↔ SOX2
evidence Significant decrease in luciferase reporter promoter activity (quantitative)
context Dual luciferase reporter assay in 293T cells
confidence 0.9 · significant: True
source_span Page 5, Para 1

Individual knockdown of Cbx4 or Cbx6 promotes MEF reprogramming and reactivates Sox2 and Nanog gene expression.

protein_pair CBX6 ↔ NANOG
evidence Significant increase in AP+ colonies and Sox2/Nanog mRNA (quantitative)
context OSKM MEF reprogramming at day 12
confidence 0.9 · significant: True
source_span Page 6, Para 1

PATHWAY CONTEXT

Pathways: PRC1 Polycomb Repressive Pathway; Somatic Cell Reprogramming; Pluripotency and ESC Differentiation Signaling

Pathway logic: USP26 deubiquitinates and stabilizes CBX4/CBX6, assembling repressive PRC1.2 complexes on Sox2 and Nanog promoters to mediate H2AK119 monoubiquitination and suppress pluripotency.

TARGET NODE	POSITION	DYSREGULATION	DEPENDENCY EVIDENCE	SPAN
USP26	upstream_regulator	Expressed highly in MEFs and differentiated ESCs to repress pluripotency gene activation	shRNA knockdown enhances iPSC reprogramming efficiency 4-fold	Page 3, Para 1
CBX4	effector	Stabilized by USP26 deubiquitination to repress pluripotency genes via PRC1.2	shRNA knockdown enhances AP+ iPSC colony formation and upregulates Sox2/Nanog	Page 6, Para 1
CBX6	effector	Stabilized post-translationally by USP26 to repress Sox2, Nanog, and Cbx7 expression	shRNA knockdown increases reprogramming efficiency and Sox2/Nanog levels	Page 6, Para 1
SOX2	transcription_factor	Repressed through promoter H2AK119 ubiquitination mediated by stabilized CBX4/CBX6-PRC1.2	Upregulated upon Usp26, Cbx4, or Cbx6 shRNA knockdown during reprogramming	Page 5, Para 1
NANOG	transcription_factor	Repressed during differentiation via USP26-promoted PRC1.2 promoter recruitment	Markedly upregulated upon depletion of Usp26, Cbx4, or Cbx6	Page 5, Para 1

Upstream regulators

- CBX7

Downstream effectors

- SOX2
- NANOG
- CBX7

Redundancy risks

- CBX4 and CBX6 act redundantly downstream of USP26 to mediate pluripotency gene repression

CONTRADICTIONS AND NEGATIVE RESULTS (3)

FINDING	CONFLICTS WITH	REASONING
USP26 does not interact with PRC1 components RING1B, RYBP, PCGF1, BMI1, KDM2B, or PHC1, nor with PRC2 components EZH1, EED, or EZH2.	False	USP26 selectively associates with canonical PRC1.2 complex members rather than broad Polycomb complexes.
USP26 catalytic inactive mutant (C304S) and deletion mutant (Δ295-312) fail to remove K48-linked polyubiquitination from CBX4 or CBX6.	False	Catalytic deubiquitinase activity and the conserved active domain are strictly required for substrate stabilization.
Combined knockdown of Usp26 with Cbx4 or Cbx6 does not produce synergistic increases in AP+ iPSC colony counts over Usp26 knockdown alone.	False	CBX4 and CBX6 operate epistatically within the same linear regulatory pathway downstream of USP26.

METHODOLOGY

Methods

- shRNA screen
- CRISPR-Cas9 knockout
- Co-immunoprecipitation
- Western blotting
- ChIP-qPCR
- RT-qPCR

Controls

- Control shRNA
- Empty vector (EV)
- Normal rabbit/mouse IgG
- USP26 catalytic mutant C304S
- USP26 deletion mutant Δ295-312

- Luciferase reporter assay
- In vitro deubiquitination assay

Instruments

- StepOne Real-Time PCR System
- Olympus 1X71S1F fluorescence microscope
- Misonix ultrasonic liquid processor

Source organisms (NCBI taxon)

10090, 9606

PDB accessions (0)

ENTITIES

Proteins

USP26, CBX4, CBX6, CBX7, RING1A, PCGF2, SOX2, NANOG, POU5F1, DPPA4

Chemicals

Doxycycline, Retinoic acid, MG132, Lactacystin, Puromycin, Polybrene, CHIR99021, PD0325901

NORMALISED IDENTIFIERS (SIDECAR; 10 RESOLVED, 0 UNRESOLVED, 0 FILTERED)

RAW NAME	GENE	UNIPROT	TIER
USP26	USP26	Q9BXU7	exact_gene
NANOG	NANOG	Q9H9S0	exact_gene
CBX4	CBX4	000257	exact_gene
CBX6	CBX6	095503	exact_gene
SOX2	SOX2	P48431	exact_gene
CBX7	CBX7	095931	exact_gene
RING1A	RING1	Q06587	uniprot_synonym
PCGF2	PCGF2	P35227	exact_gene
POU5F1	POU5F1	Q01860	exact_gene
DPPA4	DPPA4	Q7L190	exact_gene

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